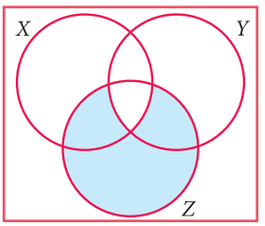
18. If $J$ = Set of three sided shapes, $K$ = Set of shapes with two equal sides and $L$ = Set of shapes with right angle, then $J \cap K \cap L$ is

(1) Set of isoceles triangles (2) Set of equilateral triangles

(3) Set of isoceles right triangles (4) Set of right angled triangles

19. The shaded region in the Venn diagram is




(1) $Z-(X \cup Y)$ (2) $(X \cup Y) \cap Z$

(3) $Z-(X \cap Y)$ (4) $Z \cup (X \cap Y)$

20. In a city, 40% people like only one fruit, 35% people like only two fruits, 20% people like all the three fruits. How many percentage of people do not like any one of the above three fruits?

(1) 5 (2) 8 (3) 10 (4) 15

## Points to Remember

- A set is a well defined collection of objects.
- Sets are represented in three forms (i) Descriptive form (ii) Set – builder form (iii) Roster form.
- If every element of $A$ is also an element of $B$, then $A$ is called a subset of $B$.
- If $A \subseteq B$ and $A \neq B$, then $A$ is a proper subset of $B$.
- The power set of the set $A$ is the set of all the subsets of $A$ and it is denoted by $P(A)$.
- The number of subsets of a set with $m$ elements is $\mathbf{2^m}$.
- The number of proper subsets of a set with $m$ elements is $\mathbf{2^m - 1}$.
- If $A \cap B = \varnothing$ then $A$ and $B$ are disjoint sets. If $A \cap B \neq \varnothing$ then $A$ and $B$ are overlapping.
- The difference of two sets $A$ and $B$ is the set of all elements in $A$ but not in $B$.
- The symmetric difference of two sets $A$ and $B$ is the union of $A$-$B$ and $B$-$A$.
- **Commutative Property**

  For any two sets $A$ and $B$,

  $A \cup B = B \cup A$ ; $A \cap B = B \cap A$



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